

Representation in $\mathbb{R}^3$ of a not yet metric - possessing Trace by an Ensemble: Curve - Evolute.

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Abstract

An enlarged concept of Curve is introduced. Not only expressed by Curvature and Torsion, but by Rotations: around the Curvature Centers, and around T, N, B of the Frenet-Serret frame.

Stationary in $\mathbb{R}^3$, not yet expressed by the Euclidean metric hence called bare, a Trace is analyzed by an ordered set of Triangles, corresponding to a set of Planes, obtaining an Ensemble: Curve + Evolute. The Triangles are exactly bounded by the Curve's and Evolute's components; a Triangulation Nest in $\mathbb{R}^3$ becomes the final Reference Space for the Ensemble. The result is an Intrinsic Description of the Ensemble.

Program CPlanes.py is enclosed. Its Layer 1 assists the Trace's intrinsic parametrization; its Layer 2 analyzes the Trace's properties.

Notation

The following Notation is for the Text.

The Program has, and explains, its own Notation.

$\{\text{Curve's Parameter}\} = \{t\}.$

Index = $\{i\}$, (a low index).

$\{\text{Curve's Parameter}\}$, $\{\text{Marks' Index}\}$ have same Direction and Orientation.

Odd index for any $\{\text{Curve Triangle}\} = \Gamma.$

Even index for any $\{\text{Evolute Triangle}\} = \Lambda.$

Same odd index for $\{\text{Curve Plane}\}$, $\{\text{Curve Triangle}\}$, Arc, Arc's Curvature Center, Arc's scalar Radius, Arc's Intermediate Mark.

Same even index for $\{\text{Evolute Plane}\}$, $\{\text{Evolute Triangle}\}$ and its unique Mark.

In a Triad a Vector Radius is expressed by two low indices: the 1st indicates the tip Mark = p_i , the 2nd indicates the root $\{\text{Curvature Center}\} = C_j$.
Vector Radius $R_{ij} = [p_i - C_j].$

Unit Vectors at an Even Mark = p_j are expressed:

incoming $T_{p_j} = T_{\rightarrow p_j}$, outgoing $T_{p_j} = T_{p_j \rightarrow}$

incoming N_{p_j} , outgoing N_{p_j}

incoming B_{p_j} , outgoing $B_{p_j} = B_{p_j \rightarrow}$

Where: incoming from the *previous* $\{\text{Curve Arc}\}$,
outgoing to the *next* $\{\text{Curve Arc}\}.$

Mark = evidenced Point, common to Trace and Curve.
Any Curve's Mark is parametrized and indexed.

The set $\{\text{Marks}\}$ is linearly ordered by the Index:

Indexed Set $\{\text{Marks}\} = \{p_i\}; p_0 \leq p_i \leq p_n$

Even Mark = EM, with an even index ; Odd Mark = OM, with an odd index.

Boundary Marks (two) of an Arc. with even indexes.

Intermediate Mark (one) of an Arc. With odd index.

Oriented Mark = $p_j = \rightarrow p_j + p_j \rightarrow$

Common Mark, for a sequence of Triangles; with even index:
 $\{ \{ \text{Curve Triangle} \}, \{ \text{Evolute Triangle} \}, \{ \text{Curve Triangle} \} \} = \{ \text{Tern} \}.$
The Common Mark is a Point of each Triangle.

Triad's Curvature Center = Arc's Curvature Center = Evolute's Point = C_i ,
where i = Triad's and Arc's odd index.

$\| \text{Distance between the generic Points } q, p \| = d_{qp} \vee \| d_{qp} \| = \| p - q \|.$

Root = p, Tip = q. Vector = $[p - q]$.

$\{ \text{Reference Marks} \} = qs = \{ q_0, q_1, q_2, q_3 \}.$

Associated to the chosen, unique subset $\{ \text{Reference Marks} \} = qs$,
the set $\{ \text{Reference Distances} \}$:

$d_{00} = \| q_0 - q_0 \|, d_{01} = \| q_1 - q_0 \|, d_{02} = \| q_2 - q_0 \|, d_{03} = \| q_3 - q_0 \|.$

$d_{12} = \| q_2 - q_1 \|, d_{13} = \| q_3 - q_1 \|, d_{23} = \| q_3 - q_2 \|;$

For the $\{ \text{Reference Marks} \}$:

$\{ \text{Triples} \} = \{ \{ q_0, q_1, q_2 \}, \{ q_1, q_0, q_3 \}, \{ q_0, q_2, q_3 \}, \{ q_2, q_1, q_3 \} \}.$

Each Triple of Reference Marks defines one $\{ \text{Base Plane} \}.$

Each Oriented Triple defines an $\{ \text{Oriented Base Plane} \}.$

Function $f(t)$. We distinguish, for a function $f(t)$, the cases:

a) The abrupt, *local* at (i) change: $f(t_{\rightarrow p_i}) \rightarrow f(t_{p_i \rightarrow}).$

b) The differential from $f(t)$ to $f(t + dt)$.

Triangles. $\{ \text{Curve Triangle} \} = \Gamma. \{ \text{Evolute Triangle} \} = \Lambda.$

Different symbols for them, but an uninterrupted Ordered Set of values of the index.

$\{ D C M \} = \{ \text{Direction Cosine Matrix} \}.$

$\{ T, N, B \} = \text{Frenet - Serret frame of Unit Vectors}.$

$\{ \text{Tern} \}_{i-1} = \text{Ensemble's sequence:}$
 $\{ \{ \text{Curve Triangle} \}_{i-2}, \{ \text{Evolute Triangle} \}_{i-1}, \{ \text{Curve Triangle} \}_i \}.$

With even index (i-1).

{ Triad } = a sequence of 3 Marks.

1 Trace

Previously the Human ascertained the Euclidean Geometry of the part of space in which the Trace lies.

Stationary being independent of $\mathbb{R}^0$ in $\mathbb{R}^{3+1}$, and not yet expressed by the Euclidean metric, a Trace, a.k.a. Shape, is given and perceived by an operating in $\mathbb{R}^{3+1}$ Human Observer. That is, a deformable Trace in $\mathbb{R}^{3+1}$ is not considered here. The considered Trace is not a Set of disjoint Points; its Points are sensibly connected into the Trace itself. Hence the Trace has *Positional Continuity*. Generally it can have other types of geometric Continuity / Discontinuity. It is not, but it shall be transformed into an { Ensemble: Curve + Evolute }. It can be either regular or not regular, open or closed, with coexisting left-handed and right-handed pieces. It is the *Datum* of the Human's analysis. In this analysis, it is not stretched, not smoothed by the Human. In the Representation by said Ensemble it is associated to Measures, Marks, an Index, a Parameter, a Reference Space.

There is a set of possible Definitions of a Trace. A preliminary extrinsic definition in $\mathbb{R}^3$ is not introduced here.

An Intrinsic Definition of the ensuing Ensemble is given by a Reference Space, that, completed at the analysis' end, is a { Triangulation Nest } in $\mathbb{R}^3$, each Triangle being stationary and stationarily connected to the other Triangles.

Trace's Resolution of its geometric features. It depends on how finely the Human samples the Trace. It is a Human's choice with physical limitations, like the Human's size in relation to the Trace, her available sensors, the smallest features of interest, the required level of detail. In general it varies from place to place in the Trace.

Precision of the Representation: it is proportional to the assumed density of Marks.

2 Marked Trace

The Human performs preliminary operations on the Trace:

She perceives on the Trace a set { not yet parametrized Points }. (2.1)

She conceives a **Direction** along the Trace: the Linear Direction. (2.2)

She chooses a specified *Orientation* for this Direction; one within two. (2.3)

To the Trace, she affixes, after pondered thoughts, an Ordered Set $\{ Marks \} = \{ Marked\ Trace's\ noticeable\ Points,\ suitably\ chosen\ for\ its\ Representation \} \subset \{ Trace's\ Points \}$. (2.4)

Hence the ensuing { Marked Trace } has a specified, indexed but not parametrized set (the Curve shall be parametrized)

Directed & oriented { Marked Trace's Evidenced Points } = { Marks }.

Sequences on a Marked Trace. An *appropriate, Sequence* of indexed Marks must be assumed, and kept carefully, since the two subsets of { Marks } : { Even Marks } = { EMs }, { Odd Marks } = { OMs }, have different functions. (2.6)

a) Sequence of Marked Trace's Marks:

$\{ \{EM\} , ..., \{EM\} , \{OM\} , \{EM\} , ..., \{EM\} \};$

here it is written: $\{ Marks \} = \{ p_0 , ..., p_{i-1} , p_i , p_{i+1} , ..., p_n \}$.

Where: $p_0 , ..., p_{i-1} , p_{i+1} , ..., p_n$ are Even Marks. (2.7)

b) Sequence of Marked Trace's Intervals:

$\{ ..., \{ p_{(i-1) \rightarrow} , ..., p_{(i)} , ..., p_{\rightarrow (i+1)} \} , ... \}$. (2.18)

Marked Trace's Intervals. If not with an angle near π , any Marked Trace's Interval with constant or nearly constant Curvature, and negligible other Rotations, must be conveniently fitted by two boundary {EMs} and one intermediate OM. (2.19)

The above expedient of assuming three Marks and defining them the only { Marks } in a marked Trace's Interval with constant Curvature, is for the entire Marked Trace.

By this expedient the necessary No. of Marks for the Trace's Representation

called Curve is reduced to a minimum. (2.20)

{ Even Marks }. The Human must define the elements of the subset { Even Marks } = {EMs} of the Marked Trace. a) In a Mark, some geometric properties change abruptly. So that to define it a distinguishing EM is mandatory. b) But not necessarily an abrupt change corresponds to an EM. In one Interval, an EM with another task: the transformation of the OM into one EM, and the creation of two new OMs. Aiming to the partition of one too long Marked Trace Interval into two shorter Intervals. (2.21)

{ Odd Mark }. In the Interior of any Marked Trace's Interval that is defined by 3 Marks only. Between the two Even Marks, one, and only one

{Intermediate Mark} = { Odd Mark } = OM.

This OM is necessary in the ensuing Curve, for the definition of: the local circular { Arc } and its Curvature. Not necessarily it is the {Arc's Medium Point}, but the precision of the analysis is enhanced if said OM is near the {Arc's Medium Point}. (2.22)

In the same Marked Trace's Interval:

{ Marked Trace's non-evidenced Points } $\neq$ { Marks }. (2.23)

Index. The set { Marks } has an ordering { Mark Index } = { i },

hence the { Marked Trace } has { Mark Index }. (2.24)

Marked Trace's { Marks } are indexed but not yet parametrized, by definition. (2.25)

Accuracy. The Representation of the { Trace } by an ensuing { Curve } is accurate if Human's attention to abrupt changes and smooth changes is applied in the transformation of the { Trace } into a { Marked Trace }. (2.26)

3 Curve

{ Curve }. A { Curve } is a Human-conceived Representation of a given { Trace }. In this study the process of Trace's Representation is:
Trace -> Marked Trace -> Curve -> Ensemble.

{Regular Curve}. Math coontains the concept of {Regular Curve}.

{Non-regular Curve}. Here, the superposed pieces of a {Non-regular Curve} are analyzed locally and separately. This is possible since the elements (Points, Arcs, Rays, Vectors, Planes, Triads, Tetrads, Terns) of each piece are mot only parametrized and indexed in $\mathbb{R}^3$, but also linearly and distinguishably along the Curve. So that the analysis of one piece ignores the other superposed pieces. (3.1)

Generic Curve. A generic Curve may have: Abrupt change Marks, finite Arcs and differential Arcs. (3.2)

{ Marks }. $\{ Marks \} = \{ Indexed Points \} = \{ Marked Trace's Evidenced Points \} = \{ Curve's Evidenced Points \}.$

{Marked Trace} and ensuing {Curve} coincide obligingly only at the Marks.

The { Marked Trace's Marks } are assumed as { Curve's Marks }. (3.3)

{Non-evidenced Points}. $\{ Marked Trace's non-evidenced Points \} \neq \{ Marks \} \neq \{ Curve's non-evidenced Points \}.$ Because:

As the name tells us: {Marked Trace non-evidenced Points} are Marked Trace's properties.

The { Curve's non-evidenced Points } lie in circular { Arcs } $\subset$ Curve ; generally not strictly along the Marked Trace.

But: each Curve's circular Arc approximates a Marked Trace's Interval. (3.4)

The Human strives to get their nearness in $\mathbb{R}^3$. (3.5)

Curve's Parameter. The Human applies the **Parameter** {t} to the Curve. Each Point (Mark and Non-evidenced Point) of the Curve gets a Parameter value. (3.6)

$\{ Curve's Points \} = \{ p_t \}.$ (3.7)

$\{ Curve's indexed \& parametrized Points \} = \{ Curve Marks \} = \{ p_i \}$ (3.8)

$\{ p_t \} \supset \{ p_i \}.$ (3.9)

Density of Curve's Marks. It is relative to the Parameter.

It was said before for the Marked Trace. Where the $\{\text{Marks}\}$ are placed tightly, the Curve varies with a sequence of short Arcs. (3.10)

An Open Curve has Even End Marks p_0, p_n . (3.11)

An Open Curve that begins with one EM and ends with one OM has an incomplete and undefined final $\{\text{Arc}\}$. (3.12)

A Closed Curve has superposed Initial and Final Even Marks p_0, p_n . (Distinct indexes) (3.13)

Representation of a Trace by a Curve. According to the Human:

1] By definition: one $\{\text{Marked Trace's Mark}\}$, and the corresponding $\{\text{Curve's Mark}\}$, have the same position in an assumed $\mathbb{R}^3$. (3.14)

2] A $\{\text{Marked Trace's Non-evidenced Point}\}$ is represented by the $\{\text{Curve's Non-evidenced Point}\}$ that is in $\mathbb{R}^3$ at the minimum distance from it. (3.15)

3] A smooth $\{\text{Transition}\} = \{\text{change of some value}\}$ in the Marked Trace must correspond to a smooth $\{\text{Transition}\}$ in the Curve. A non-smooth Transition at a Marked Trace's EM must correspond to a non-smooth Transition at the Curve's same indexed EM. (3.16)

$\{\text{Pair of Marks in sequence}\}$. It is a Curve's component, not of the Evolute. By $\{t\}$, it may define a Distance $= \Delta^1 t$. (3.17)

$\{\text{Triad}\}$. A $\{\text{Triad}\}$ is a Curve's component, not of the Evolute.

It is defined by a subset of 3 Marks of the Marked Trace: one in the Triad incoming EM, one intermediate OM, one from the Triad outgoing EM, and their $\{\text{Interval Interior}\}$, i. e. $\{\text{Arc Interior}\}$. Three Indexes in the Sequence $\{\text{Marks}\}$ correspond to them. One $\{\text{Triad Triangle}\} \subset \{\text{Triad Plane}\}$, one $\{\text{Triad Arc}\}$ in it. And their properties. One single $\{\text{Triad}\}$ may define a $\Delta^2 t$, Curvature, but not Torsion etc. (3.18)

In the Triad's Triangle: the Arc's Curvature, and the Arc's Curvature Center that is a Triad's property and is ascribed as an Evolute's Point. (3.19)

The introduction of $\{\text{Triad}\}$ is due to the Points' principle, here Marks' principle too: *Always 3 distinct, not aligned Points define a Plane and in it an Arc of Circle. If the Triad has 3 distinct, aligned Points, a straight Arc is*

defined, but not a Plane. (3.20)

{ Tetrad }. A { Tetrad } is a sequence of four Points; eventually of four Marks. One Tetrad may define a $\Delta^3 t$ and Torsion. (See the subsection "Differential Vectors on the Curve"). (3.22)

{ Triads }. The ordered set { Marked Trace's Marks } is associated to the ordered set { Marked Trace's Triads }. The ordered set { Curve's Marks } is associated to the ordered set { Curve's Triads }. (3.23)

{ Arc }. The { Arc } of a Triad is an ordered subset of Curve's { Points }, comprehensive of: 3 evidenced Points (no less, no more) called Marks (one Triad), and other non-evidenced Points. It is a Curve's Interval with a Marks-defined, constant Curvature (not Torsion, etc.: 3 Marks not sufficient).. (3.24)

In the Arc Interior, the geometric properties change smoothly, by definition. If not so, the set { Marks } must be reorganized. (3.25)

Generic Arc. $\{ \text{Arc} \}_i = A_i \supset \{ p_{(i-1) \rightarrow}, \dots, p_i, \dots, p_{\rightarrow(i+1)} \} = \{ \text{Outgoing from } p_{i-1}, \dots, p_i, \dots, \text{Incoming into } p_{i+1} \}.$

$\{ \text{OM} \} = p_i. \{ \text{EMs} \} = \{ p_{(i-1) \rightarrow}, p_{\rightarrow(i+1)} \}.$ Odd i (3.26)

Arc - length. In general, the Curve's { Arcs } do not have equal lengths. (3.27)

Arc's Frenet-Serret field. Each Arc A_i has its $\{ T, N, B \}_{A_i}$ Frenet-Serret field, bar a straight Arc A_j having its T_{A_j} field only. (3.28)

Incoming in $\text{Int } A_i$, hence Outgoing from p_{i-1} : $\{ T, N, B \}_{p_{(i-1) \rightarrow}}.$

Outgoing from $\text{Int } A_i$, hence Incoming into p_{i+1} : $\{ T, N, B \}_{p_{\rightarrow(i+1)}}.$

Outgoing from p_{i-1} that is the { Initial Mark } of A_i : $\{ T, N, B \}_{p_{(i-1) \rightarrow}}.$

Incoming into p_{i+1} that is the { Final Mark } of A_i : $\{ T, N, B \}_{p_{\rightarrow(i+1)}}.$ (3.29)

In A_i . For a non-straight Arc A_i . Changing Point in the Interior of A_i , for definition of Triad: { T }, { N }, change smoothly; { B } changes its { Point of application } only, keeping constant Direction & Orientation. The field $\{ B \}_{A_i}$ is perpendicular to A_i ; all its elements B_p are parallel. (3.30)

Straight Arc. A {Straight Arc} is a 3 - Marks - defined straight Segment.
 Its Curvature Center is at ∞ , not in a defined Plane.
 Its Boundary Vector Radii are parallel, not in a defined Plane.
 Defined { T }, undefined { N }, { B }. (3.31)

Degenerate Arc # 1, An { Arc } with one EM and one OM only, or two EMs and no OM. It is incomplete and undefined. A revision of the set { Marks } is necessary. (3.32)

Degenerate Arc # 2. An { Arc } with more than 3 Marks, in general without uniform curvature. For this reason it must be decomposed. (3.33)

Differential Triad. A Differential Triad has the usual 3 Marks and a tiny Arc of 2, generally not aligned, Differential Intervals.

And as usual, also for a Differential Triad the two Boundary {Even Marks} = EMs and the Intermediate {Odd Mark} = OM define the Triad and the Index order. (3.34)

In a Differential Triad there is the advantage of defining locally the Curve uniquely by an easy to define Curvature, not by a set of elementary Rotations. (3.35)

Sequence of Differential Triads. Applying a big No. of Marks in a given Marked Trace Interval, and in the corresponding Curve Interval, by a Sequence of Differential Triads, the fidelity of the local Representation is enhanced.

The complication of the Trace induces a big No. of Differential Triads in Sequence (obviously this is an inconvenience). (3.36)

Differential Arc. Say $\{\text{Arc}\}_i = A_i \supset \{p_{(i-1) \rightarrow}, \dots, p_i, \dots p_{\rightarrow(i+1)}\}$.
 Odd i, with a Pair of Differential, straight Intervals. In general, non-null Curvature, hence non-straight Arc, even if of tiny length. (3.37)

Differential Vectors on the Curve and their Points of application.

(An impractical symbolism is used to enhance understanding.)

The sum of Vectors with the same origin is considered allowable. (3.38)

It is well known that 2 Points are necessary to get a 1st differential; 3 Points

to get a 2nd differential; 4 Points to get a 3rd differential (3.39)

A 3rd Differential on the Curve. Let us suppose on the Curve 4 Points: t_1, t_2, t_3, t_4 and the Differential Vectors $\Delta^1 t_{1 \rightarrow 2} = [t_2 - t_1]$ applied at t_1 , $\Delta^1 t_{2 \rightarrow 3} = [t_3 - t_2]$ applied at t_2 , $\Delta^1 t_{3 \rightarrow 4} = [t_4 - t_3]$ applied at t_3 . (3.40)

They are assumed straight Differential Vectors because

$$\|t_2 - t_1\|, \|t_3 - t_2\|, \|t_4 - t_3\|$$

are very little in comparison to the local Curvature Radius. (3.41)

Changing sign, the Point of application changes; for example:

$\Delta^1 t_{1 \rightarrow 2}$ is applied at t_1 ,

$-\Delta^1 t_{1 \rightarrow 2} = \Delta^1 t_{2 \rightarrow 1}$ is applied at t_2 and has the tip at t_1 .

A longitudinal translation $\Delta^1 t_{1 \rightarrow 2} = \Delta^1 t_{2 \rightarrow a}$, yields: $\Delta^1 t_{2 \rightarrow a}$, that is applied at t_2 and has the tip at a .

At t_2 : $\Delta^1 t_{2 \rightarrow a}$ is vectorially added to $\Delta^1 t_{3 \rightarrow 2}$, getting $\Delta^2 t_{3 \rightarrow a}$, applied at t_3 . ($\Delta^2 t_{3 \rightarrow a}$ has the same size and direction, opposed orientation of $\Delta^2 t_{2 \rightarrow b} = \Delta^1 t_{2 \rightarrow 1} + \Delta^1 t_{2 \rightarrow 3}$, that is applied at t_2). (3.42)

By the parallelogram construction of two Vectors, both with origin at t_3 , we get: $\Delta^2 t_{3 \rightarrow c} = \Delta^1 t_{3 \rightarrow 2} + \Delta^1 t_{3 \rightarrow 4}$ that we say applied at t_3 , not to t_2 . (3.43)

$\Delta^3 t_{3 \rightarrow d} = \Delta^2 t_{3 \rightarrow a} + \Delta^2 t_{3 \rightarrow c}$, that we say applied at t_3 , where there is the vectorial sum. (3.44)

Derivatives. By the above we get the Derivatives:

$$\Delta^1 t / \Delta t \quad ; \quad \Delta^2 t / 2 \Delta t^2 \quad ; \quad \Delta^3 t / 6 \Delta t^3. \quad (3.45)$$

{Curve Plane}. To each {Triad} and its corresponding {Arc}, necessarily a Plane in $\mathbb{R}^3$ corresponds. Since it contains elements of the Curve, it is named {Curve Plane}. (3.46)

$$\{\text{Curve Plane}\}_i \supset \{\text{Curve Triangle}\}_i = \Gamma_i. \quad \text{Odd } i. \quad (3.47)$$

{Curve Triangle}. say Γ_i . Γ_i contains: the Triad, the Arc = A_i with all its {Points} and within them its {Triad's Marks} = $\{p_{(i-1) \rightarrow}, p_i, p_{\rightarrow(i+1)}\}$, the local {Curve Center} $\}_i = \{\text{Arc's Curvature Center}\}_i = \{\text{Evolute Point}\}_i$

$= C_i$ that is associated to the Rotation (see a following subsection) θB_{C_i} , (around B_{C_i} , not around B_{p_i}) the {Scalar Radius} $= r_i$, common to all Points of A_i , the Vector Radii, one for each Point of A_i , within them the {Initial Vector Radius} $= R_{(i-1)(i)} = [p_{i-1} - C_i]$ and the {Final Vector Radius} $= R_{(i+1)(i)} = [p_{i+1} - C_i]$. (3.48)

{Curve Triangle}'s relevants Points:

one OM, two EMs, one {Curvature Center}. (3.49)

Curve's {Initial Curve Triangle}. The Open $\vee$ Closed Curve's {Initial Curve Triangle} be Γ_1 ; OM $= p_1$, {EMs} $= \{ p_{0 \rightarrow}, p_{\rightarrow 2} \}$, {Initial Triad} $_1 = \{ p_{0 \rightarrow}, p_1, p_{\rightarrow 2} \}$, {Initial Curvature Center} $= C_1$. (3.50)

Curve's {Final Curve Triangle}. For an Open $\vee$ Closed Curve, the {Final Curve Triangle} be Γ_{n-1} , OM $= p_{(n-1)}$, {EMs} $= \{ p_{(n-2) \rightarrow}, p_{\rightarrow n} \}$, {Final Triad} $_{n-1} = \{ p_{(n-2) \rightarrow}, p_{n-1}, p_{\rightarrow n} \}$, {Final Curvature Center} $= C_{n-1}$. (3.51)

For a Closed Curve: $p_{0 \rightarrow} = p_{\rightarrow n}$. (3.52)

{Curve Triangle} with a straight Arc..

If the {Arc} is a nonnull straight Segment, that is defined by its OM and the two EM, then: {Curvature Center} $= C = \infty$. Scalar Radius $= \infty$. Initial Vector Radius, Final Vector Radius are extant, parallel but undefined. Arc's Length = Segment's Length. Field {T} of the straight Arc's Points with constant Direction and Orientation; undefined {N}, undefined {B}. (3.53)

Odd Index. Same Odd Index for the associated items: Triad, OM, Arc, Curvature Center of this Arc, Curve Plane, Curve Triangle. (3.54)

Ordered set { Curve Triangles }. They meet in pairs, and with an intermediate { Evolute Triangle }, at the Even Marks.

Say: $\{ \dots, \Gamma_{i-2}, \Lambda_{i-1}, \Gamma_i, \Lambda_{i+1}, \Gamma_{i+2}, \dots \}$. With:

$\Gamma_{i-2} \supset \{ p_{i-3}, p_{i-2}, p_{i-1} \}$, $\Gamma_i \supset \{ p_{i-1}, p_i, p_{i+1} \}$, $\Gamma_{i+2} \supset \{ p_{i+1}, p_{i+2}, p_{i+3} \}$.
The even Mark p_{i-1} is common to the odd { Curve Triangles }: Γ_{i-2}, Γ_i .
The even Mark p_{i+1} is common to the odd { Curve Triangles }: Γ_i, Γ_{i+2} .
In general: $R_{(i-1)(i-2)} \neq R_{(i-1)(i)}$ and $R_{(i+1)(i)} \neq R_{(i+1)(i+2)}$, that implies,

in the Evolute: $C_{i-2} \neq C_i$ and $C_i \neq C_{i+2}$. (3.55)

4 Evolute

The { Evolute } is analyzed by the ordered set { Evolute Planes }.

{Evolute Plane}. An { Evolute Plane } is defined by the {Previous Triad's Final Vector Radius} and the {Next Triad's Initial Vector Radius}, which have their tips at a common EM, that is a {Curve's Mark} and an {Evolute Plane's Point}, and two distinct roots: {Previous Triad's Curvature Center}, {Next Triad's Curvature Center}; they are {Evolute's Points} and also {Curve Planes' Points}. (4.1)

Any {Evolute Plane}, say {Evolute Plane} $_{i+1}$, contains one {Evolute Triangle}, say Λ_{i+1} . (4.2)

{Evolute Triangle}.

An {Evolute Triangle}, say Λ_{i+1} , has a { Tripod } = { three main Points }:
EM = $p_{(i+1)}$; {Previous Triad's Curvature Center} = C_i ; {Next Triad Curvature Center} = C_{i+2} . They are {Evolute Triangle}'s Points and also {Curve Triangle}'s Points. (4.3)

An {Evolute Triangle} is between two {Curve Triangles}, in both cases: Open Curve, Closed Curve.

Say: the {Evolute Triangle} = Λ_{i+1} is between Γ_i , Γ_{i+2} . (4.4)

{Closed Curve's additional Evolute Triangle. }

{ Additional Evolute Triangle } = $\Lambda_0 \supset \{ p_0 = p_n, C_{n-1}, C_1 \}$.

Superposed p_0, p_n . (4.5)

A possible anomaly. By definition, an { Evolute Triangle } has an Arc of one {EM} only. If an { Evolute Triangle }, say Λ_{i-1} , has an {Arc} = A_{i-1} of a non-null length, then a rearranging of the Curve is required. (4.6)

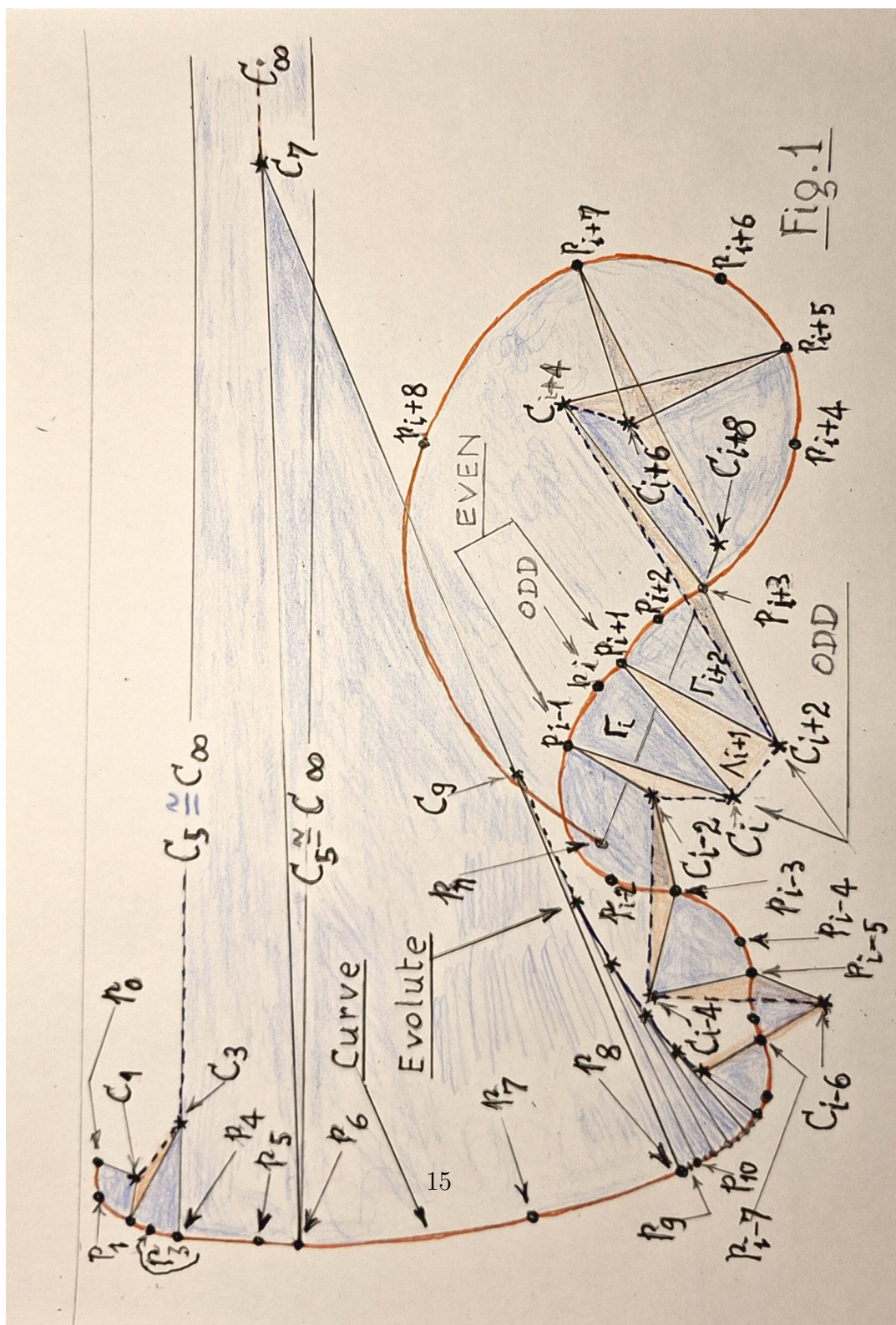
Ordered set { Evolute Triangles }. Say:

$\dots, \Lambda_{i-5}, \Lambda_{i-3}, \Lambda_{i-1}, \Lambda_{i+1}, \Lambda_{i+3}, \Lambda_{i+5}, \dots$ (4.7)

The { Evolute Triangles } meet in Pairs at the Curvature Centers of the

interposed { Curve Triangles }. (4.8)

5 {Ensemble} = {Curve & Evolute}.



Intrinsic Representation of a Trace by the {Ensemble}.

An { Ensemble } = { Curve , Evolute } can represent a Trace.

In the Representation, continuities and discontinuities of the Trace must be kept in the Ensemble. (5.1)

Sequences. In said Representation: a) for the Curve, a Sequence of Triads, three Marks for each Triad; b) for the Evolute, a Sequence of Tripods, three Points for each Tripod, i.e. two Curvature Centers and one Even Mark.

Even Marks are Points of the Curve and of the { Evolute Planes }.

Curvature Centers are Points of the Evolute and of the { Curve Planes }.

(That is, some sort of symmetry: Curve, Evolute). (5.2)

Ordered Aggregates in the Ensemble: { Terns }, { Triplets }. (5.3)

{ Tern }. In the Ensemble, a { Tern } is any oriented aggregate:
 $\{ \{ \text{Previous Curve Triangle} \} , \{ \text{Evolute Triangle} \} , \{ \text{Next Curve Triangle} \} \}$.

Where Previous, Next are with reference to the { Evolute Triangle }.

Only one Mark, that is an EM, in common to the Tern's three Triangles.

Generally a Tern is in $\mathbb{R}^3$. Say:

$$\begin{aligned} \{ \text{Tern} \}_{i-1} &= \{ \{ \text{Curve Triangle} \}_{i-2} , \{ \text{Evolute Triangle} \}_{i-1} , \{ \text{Curve Triangle} \}_i \} \\ &= \{ \Gamma_{i-2} , \Lambda_{i-1} , \Gamma_i \}. \quad \text{With: } \Gamma_{i-2} \supset C_{i-2} . \Lambda_{i-1} \supset \{ C_{i-2} , C_i \} . \\ &\Gamma_i \supset C_i . \text{ Generally: } C_{i-2} \neq C_i . \text{ For the common Even Mark } = p_{i-1} : \\ &\Gamma_{i-2} \supset p_{\rightarrow(i-1)} ; \Lambda_{i-1} \supset p_{i-1} ; \Gamma_i \supset p_{(i-1)\rightarrow} . \end{aligned} \quad (5.4)$$

A Rotation along B in any single { Curve Triangle }.

{ Terns } are introduced to evaluate Rotations in the entire $\mathbb{R}^3$. (5.5)

Tern's properties. In a Tern, the { Previous Curve Triangle } yields its intrinsic properties. The { Next Curve Triangle } yields its intrinsic properties. Their intermediate { Evolute Triangle } yields *the two { Curve Triangles }' relative properties.* (5.6)

In each Tern, a sequence:

{ Curve's Previous Arc }, EM, { Curve's Next Arc }.

Note. One EM between two Arcs avoids the following inconveniences.

1] The partial superposition of two non-pointlike Arc Intervals, and the en-

suing necessary of getting, from these two Arc Intervals, a necessarily unique, chosen Curve's Interval.

2] The impossible representation of Abrupt Rotations in the Curve. (5.7)

{ Terns }. The ordered set { Terns } is used to analyze the entire Ensemble. (5.8)

Two main subsets: { Abrupt Terns }, { Differential Terns }. (5.9)

{ Abrupt Tern } i.e. Jump.

The { Evolute Triangle }, say Λ_{i-1} , by definition having one Mark i.e. { EM } = p_{i-1} only, not a Curve's Interval, expresses a Jump at said

$p_{(i-1)} = p_{\rightarrow(i-1)} \cup p_{(i-1) \rightarrow}$ if
 $T_{\rightarrow p_{(i-1)}} \neq T_{p_{(i-1) \rightarrow}} \vee N_{\rightarrow p_{(i-1)}} \neq N_{p_{(i-1) \rightarrow}} \vee B_{\rightarrow p_{(i-1)}} \neq B_{p_{(i-1) \rightarrow}}, \vee$ a composition of them. That implies: $C_{i-2} \neq C_i$. There is not a continuous { Evolute's Arc }. i. e. there are undefined Centers in the Interior of the Interval $[C_i - C_{i-2}]$. (5.10)

Tern Cases. For the Tern $\{ \Gamma_i, \Lambda_{i+1}, \Gamma_{i+2} \}$. (For these cases, the Frenet-Serret frames shall be analyzed in a following subsection.)

Case 1. { Curve Triangles } $\Gamma_i \neq \Gamma_{i+2}$.

{ Evolute Triangle } = $\Lambda_{i+1} \supset \{ p_{i+1}, C_i, C_{i+2}, R_{(i+1)(i)}, R_{(i+1)(i+2)} \}$.

{ Curvature Centers } = $\{ C_i, C_{i+2} \}$; { EM } = p_{i+1} ; For Λ_{i+1} :

Initial Vector Ray = $R_{(i+1)(i)} = [p_{i+1} - C_i]$;

Final Vector Ray = $R_{(i+1)(i+2)} = [p_{i+1} - C_{i+2}]$. (5.11)

Case 2. A special case. { Curve Triangles } $\Gamma_i \neq \Gamma_{i+2}$. They are non-coplanar, and with curvatures having same values.

The { Evolute Triangle } = Λ_{i+1} is an **Edge** between them. On this Edge:

$R_{(i+1)(i)} = R_{(i+1)(i+2)}$; $\| R_{(i+1)(i)} \| = \| R_{(i+1)(i+2)} \|$; $C_i = C_{i+2}$. (5.12)

Case 3. A special case. { Curve Triangles } $\Gamma_i \neq \Gamma_{i+2}$. They are non-coplanar, and with different curvatures.

The { Evolute Triangle } = Λ_{i+1} is an **Edge** between them. On this Edge:

$R_{(i+1)(i)} \neq R_{(i+1)(i+2)}$; $\| R_{(i+1)(i)} \| \neq \| R_{(i+1)(i+2)} \|$; $C_i \neq C_{i+2}$. (5.13)

Case 4. A special case. { Curve Triangles } $\Gamma_i \neq \Gamma_{i+2}$. They are coplanar, but with different curvatures.

The {Evolute Triangle} = Λ_{i+1} is a **Segment** between them. On it:

$$R_{(i+1)(i)} \neq R_{(i+1)(i+2)} ; \| R_{(i+1)(i)} \| \neq \| R_{(i+1)(i+2)} \| ; C_i \neq C_{i+2} . \quad (5.14)$$

Case 5. A special case. Coplanar Triangles Γ_i , Λ_{i+1} , Γ_{i+2} . $\Gamma_i \neq \Gamma_{i+2}$. The { Curve Triangles } have equal scalar Radii, different Vector Radii, different Curvature Centers. Between them, the {Evolute Triangle} = Λ_{i+1} has:
 $R_{(i+1)(i)} \neq R_{(i+1)(i+2)} ; \| R_{(i+1)(i)} \| = \| R_{(i+1)(i+2)} \| ; C_i \neq C_{i+2} . \quad (5.15)$

Differential Triad.

In Math, a { **Differential Interval between two Points** } has a tiny value and can be assumed straight with reference to the local Scalar Radius.

This is not the case of a tiny Interval along a Trace; if not a { Pair of Points } is considered, but a { **Differential Triad of Marks** }, and this { Triad of Marks } implies a {Differential Curve Plane}, a { Differential Curve Triangle }, a { Differential Curve Arc } with generally non-null curvature. (5.16)

{ Differential Tern }. A { Differential Tern } has $0 \vee 1 \vee 2$ { Differential Curve Triangles } and, among them, one { Differential Evolute Triangle }.

For any { Differential Tern } it is admissible to assume in the Evolute a small, straight by no further knowledge

$$\{ \text{Evolute's Arc} \} = [C_{\text{Next Curve Arc}} - C_{\text{Previous Curve Arc}}] . \quad (5.17)$$

Sequence { Differential Terns } corresponding to an Ensemble's Interval.

The necessity that an Interval of the Ensemble be a Sequence of 1st Differential Subintervals is the strong and smooth variation of Curvature in the corresponding Trace. In this Ensemble's Interval, the Sequence { Differential Terns } is at the Human's discretion. A good Representation of this Interval implies a high density of Curve's Marks. Unfortunately, a high density of Marks corresponds to a big set of measured distances dv (in the program). (5.18)

That is, the Human introduces in the Ensemble a numerous Sequence of tiny { Curve Triangles }, each one with a very short Arc, and of tiny { Evolute Triangles }, each one with two very near Centers of curvature. (5.19)

In any {Differential Curve Triangle} the { Differential Triad } has one tiny

Arc with a specified Curvature Center. Hence this Arc, although tiny, is not a Straight Segment. But two Differential Arcs compose this tiny Arc; they can be assumed straight, since Scalar Radius $\gg$ Differential Arc's Length. (5.20)

What is the consequence of an eventual badly approximate representation of this Ensemble Interval? *It is a local imprecise correspondence of this Ensemble's Interval with that of a more precisely measured Ensemble's Interval.* (5.21)

In a Sequence { Differential Terns }:

a) Initial Differential Tern. The supposedly given {Initial Curve Triangle} has a finite Arc-length. It is followed by a {Differential Evolute Triangle} with two very near Curvature Centers. It is followed by a {Differential Curve Triangle} with a tiny Arc-length. (5.22)

b) A numerous set { Differential Intermediate Terns } may occur.

A { Differential Intermediate Tern } is composed by:

{Previous Curve Triangle} with tiny Arc-length;
 {Differential Evolute Triangle} with two very near Curvature Centers;
 {Next Curve Triangle} with tiny Arc-length. (5.23)

c) Final Differential Tern. The {Previous Differential Curve Triangle} has a tiny Arc-length. It is followed by a {Differential Evolute Triangle} with very near Curvature Centers. It is followed by the {Final Curve Triangle} with a finite Arc-length. (5.24)

An { Ensemble's Interval } corresponds to the above a - b - c. (5.25)

Rotation. A "Rotation" is short for "Rotation angle" .

It is around a specified Axis that is across a specified Point.

In a {Curve Triangle}, of the Outgoing Vector Radius with reference to the Incoming Vector Radius.

In a { Tern }, of the {Next Curve Triangle} with reference to the {Previous Curve Triangle}. (5.26)

Rotation classes.

Class by { t }.

a] Abrupt Rotations. Implying a (t) value. (5.27)

b] Differential Rotations. Implying a tiny Interval $\Delta t \vee \delta t$ (5.28)

c] A definite Rotation, implying a finite Interval. (5.29)

Class by Axis.

d] Around a specified Axis across the $\{Curvature\ Center\}$ of the $\{Curve\ Triangle\}$. (5.30).

e] Around a specified Axis across the EM of the reference Previous Curve Triangle of a $\{Tern\}$. (5.31)

$\{T, N, B\}$ and $\{DCM\}$. Any Rotation in the field $\{T, N, B\}$ is expressed by the $\{Direction\ Cosine\ Matrix\} = \{DCM\}$. $\{DCM\}$ encodes $\{T, N, B\}$ consistently. (5.32)

Basic Rotations within a $\{Curve\ Triangle\}$.

Case c1. In a $\{Curve\ Triangle\}$, say Γ_i , of which the Interior is considered. The constant Curvature that is associated to the scalar Radius and to the Vector Radii, is also associated to a Rotation angle, around the, constant in Γ_i , Unit Binormal Vector $\{B\}_{C_i}$ at the constant Curvature Center C_i : of the $\{Incoming\ to\ the\ Arc's\ Final\ Mark\ Vector\ Radius\}$ relative to the $\{Outgoing\ from\ the\ Arc's\ Initial\ Mark\ Vector\ Radius\}$. It is the geometric Angle at C_i of the $\{Curve\ Triangle\} = \Gamma_i$. Obviously, it is not an Abrupt Rotation.

$\{Rotation\ Angle\} = \nu B_{C_i}$, in the
 $\left(Int\ Arc \right)_i = \left(p_{i-1} \rightarrow, \dots, p_i, \dots, \rightarrow p_{i+1} \right)$,

of the, incoming to the Arc's Final Mark, $T_{\rightarrow p_{i+1}}$
relative to the, outgoing from the Arc's Initial Mark, $T_{p_{i-1} \rightarrow}$;

of the, incoming to the Arc's Final Mark, $N_{\rightarrow p_{i+1}}$
relative to the, outgoing from the Arc's Initial Mark, $N_{p_{i-1} \rightarrow}$. (5.33)

Abrupt Rotations at an EM.

A definition: *Abrupt Rotations at an EM occur sharply, at the Parameter Value (t) of the EM.* Abrupt Rotations are not Differential Rotations.

An Abrupt Rotation at an EM may imply, but not necessarily: 1] Change of Curvature Center. 2] Change: previous Arc's Scalar Radius, next Arc's Scalar Radius. 3] Change: {Incoming Vector Radius}, {Outgoing Vector Radius}. 4] Change: Incoming Frenet-Serret frame, Outgoing Frenet-Serret frame.

Possible Abrupt Rotations: 4.1] around $\{T\}_{\rightarrow EM}$; 4.2] around $\{N\}_{\rightarrow EM}$ i.e. a Torsion; 4.3] around $\{B\}_{\rightarrow EM}$ (please note: not around $\{B\}_C$); 4.4] a composition of these Elemental Rotations. (5.34)

The Curve has Parameter $\{t\}$; the Evolute has not an additional Parameter. In an Abrupt Rotation at an EM, which has a specified (t) , the Evolute's Interval is *specified* only by its two end Points. (5.35)

$\{T, N, B\}_{EM}$. In a Tern, say $\{\Gamma_i, \Lambda_{i+1}, \Gamma_{i+2}\}$ with its central EM = $p_{(i+1)} = \rightarrow p_{(i+1)} \cup p_{(i+1)} \rightarrow$, an Abrupt Rotation implies: incoming into EM, from the {Previous Curve Triangle} $\{T, N, B\}_{\rightarrow p_{(i+1)}} \neq$ outgoing from EM to the {Next Curve Triangle} $\{T, N, B\}_{p_{(i+1)} \rightarrow}$. (5.36)

$\{T, N, B\}_{EM}$ and $\{DCM\}_{EM}$. At said EM = $p_{(i+1)}$, the orientation of the outgoing $\{T, N, B\}_{p_{(i+1)} \rightarrow}$ relative to the incoming $\{T, N, B\}_{\rightarrow p_{(i+1)}}$ is defined by 9 Angles, equivalently by the {Direction Cosine Matrix} $\}_{p_{(i+1)}} = \{DCM\}_{p_{(i+1)}}$. (*Not only by Curvature and Torsion*). (5.37)

Basic Rotations around an Axis at an EM, hence Abrupt Rotations. Say at the EM = p_{i+1} of the Tern $\{\Gamma_i, \Lambda_{i+1}, \Gamma_{i+2}\}$. Basic means around $T \vee N \vee B$. (5.38)

Case d1. $\{\text{Tern}\} = \{\Gamma_i, \Lambda_{i+1}, \Gamma_{i+2}\}$. A special case.

{Evolute Triangle} = $\Lambda_{i+1} \perp \Gamma_i, \Lambda_{i+1} \perp \Gamma_{i+2}$. That is $\Lambda_{i+1} \perp T_{\rightarrow p_{(i+1)}}$.

Λ_{i+1} 's boundary $\supset R_{(i+1)(i)}, R_{(i+1)(i+2)}$ with $\|R_{(i+1)(i)}\| = \|R_{(i+1)(i+2)}\|$; their tips at p_{i+1} ; C_i root of $R_{(i+1)(i)}$, C_{i+2} root of $R_{(i+1)(i+2)}$. $C_i \neq C_{i+2}$.

$T_{p_{(i+1)}} = T_{\rightarrow p_{(i+1)}} \cup T_{p_{(i+1)} \rightarrow} = \text{incoming } T_{p_{(i+1)}} \cup \text{outgoing } T_{p_{(i+1)}}$.
 $T_{p_{(i+1)}} \perp \Lambda_{i+1}$.

At $p_{(i+1)}$ around $T_{\rightarrow p_{(i+1)}}$: {Rotation Angle} = $\theta T_{p_{(i+1)}}$.

θ is the geometric Angle in Λ_{i+1} at EM = $p_{(i+1)}$ of Γ_{i+2} relative to Γ_i ; visibly in the Curve of A_{i+2} relative to A_i .

θ of $N_{p_{(i+1)} \rightarrow}$ relative to $N_{\rightarrow p_{(i+1)}}$; θ of $B_{p_{(i+1)} \rightarrow}$ relative to $B_{\rightarrow p_{(i+1)}}$. (5.39)

Case d2. Let Λ_{i+1} be an Edge between the non-coplanar Γ_i and Γ_{i+2} ; this Edge contains $R_{(i+1)(i)} \neq R_{(i+1)(i+2)} ; p_{i+1} ; C_i ; C_{i+2} ; C_i \neq C_{i+2}$.

A Rotation around the Unit Vector T_{C_i} is not for this case: its result should be a cut of the Curve at p_{i+1} and thr destruction of the Edge. (5.40)

Case d3. $\{\text{Tern}\} = \{\Gamma_i, \Lambda_{i+1}, \Gamma_{i+2}\}$. A special case.

The $\{\text{Evolute Triangle}\} = \Lambda_{i+1}$ is an $\{\text{Edge}\}$ between the non-coplanar $\{\text{Curve Triangles}\}$: Γ_i, Γ_{i+2} .

Λ_{i+1} contains: $R_{(i+1)(i)} \neq R_{(i+1)(i+2)} ; p_{i+1} ; C_i, C_{i+2} ; C_i \neq C_{i+2}$.

Null angle at p_{i+1} of the degenerate $\{\text{Evolute Triangle}\}$.

On this Edge, parallel Unit Vectors: $N_{\rightarrow p_{(i+1)}}, N_{p_{(i+1)} \rightarrow}$.

The reference $N_{\rightarrow p_{i+1}}$ is across p_{i+1} and also across C_i .

$\{\text{Rotation Angle}\} = \tau N_{\rightarrow p_{i+1}} = \text{Abrupt Torsion}$;

τ of Γ_{i+2} with reference to Γ_i .

At p_{i+1} : τ of $T_{p_{i+1} \rightarrow}$ re $T_{\rightarrow p_{i+1}}$; of $B_{p_{i+1} \rightarrow}$ re $B_{\rightarrow p_{i+1}}$. (5.41)

Case d4. Let Λ_{i+1} be special: an Edge between the non-coplanar Γ_i and Γ_{i+2} ; this Edge contains $R_{(i+1)(i)} \neq R_{(i+1)(i+2)} ; p_{i+1} ; C_i ; C_{i+2} ; C_i \neq C_{i+2}$; parallel Unit Vectors $N_{\rightarrow p_{(i+1)}}, N_{p_{(i+1)} \rightarrow}$.

$\{\text{Rotation Angle}\} = \text{Abrupt Torsion} = \tau N_{C_i}$.

This case is equivalent to case d3; it has a different Point of application of Unit Vector N , that is C_i . (5.42)

Case d5. $\{\text{Tern}\} = \{\Gamma_i, \Lambda_{i+1}, \Gamma_{i+2}\}$. A special case.

The Tern's three Triangles are in the same Plane $\perp B_{p_{i+1}}$.

The $\{\text{Evolute Triangle}\} = \Lambda_{i+1}$ contains:

EM = p_{i+1} ;

Curvature Centers C_i, C_{i+2} ;

Vector Radii $R_{(i+1)(i)}, R_{(i+1)(i+2)}$; in general $\|R_{(i+1)(i)}\| \neq \|R_{(i+1)(i+2)}\|$.

At p_{i+1} : $\{\text{Rotation Angle}\} = \psi B_{p_{i+1}}$.

ψ is the geometric Angle of Λ_{i+1} at its vertex $EM = p_{i+1}$.

ψ around $B_{p_{i+1}}$ at $EM = p_{i+1}$ and not around B_{C_i} at C_i of Γ_i .

ψ of $T_{p_{i+1} \rightarrow}$ re $T_{\rightarrow p_{i+1}}$; ψ of $N_{p_{i+1} \rightarrow}$ re $N_{\rightarrow p_{i+1}}$. (5.43)

Case d6. $\{\text{Tern}\} = \{\Gamma_i, \Lambda_{i+1}, \Gamma_{i+2}\}$. It is a special case: the two $\{\text{Curve Triangles}\}$ and the $\{\text{Evolute Triangle}\}$ are coplanar and $\perp B_{C_i}$. Λ_{i+1} contains: $EM = p_{i+1}$; C_i, C_{i+2} ; $R_{(i+1)(i)}$; $R_{(i+1)(i+2)}$.

A Rotation around the Unit Vector B_{C_i} is not considered because its result is a cut of the Curve at p_{i+1} and its destruction. (5.44)

Differential Rotations in a Sequence $\{\text{Differential Terns}\}$.

A simple, sufficiently smooth example.

Let us consider the Sequence:

$\{\dots, \Gamma_{i-2}, \Lambda_{i-1}, \Gamma_i, \Lambda_{i+1}, \Gamma_{i+2}, \Lambda_{i+3}, \Gamma_{i+4}, \Lambda_{i+5}, \Gamma_{i+6}, \dots\}$.

Where, in general, the $\{\text{Curve Triangles}\}$ are non-coplanar and pairwise hinged along Segments.

Finite $A_{i-2} \subset \Gamma_{i-2}$.

Tiny $A_i \subset \Gamma_i, A_{i+2} \subset \Gamma_{i+2}, A_{i+4} \subset \Gamma_{i+4}$.

Finite $A_{i+6} \subset \Gamma_{i+6}$.

To understand the case, in the above Sequence let us consider the Tern $\{\Gamma_i, \Lambda_{i+1}, \Gamma_{i+2}\}$.

The $\{\text{Evolute Triangle}\} = \Lambda_{i+1}$ is the Segment containing: Vector Ray $R_{(i+1)(i)}$, Vector Ray $R_{(i+1)(i+2)}$, p_{i+1} , $C_i, C_{i+2}, N_{(i+1)(i)}, N_{(i+1)(i+2)}$. (5.45)

$\{\text{Triplet}\} = \{\{\text{Evolute Triangle}\}, \{\text{Curve Triangle}\}, \{\text{Evolute Triangle}\}\}$. (5.46)

$\{\text{Triplets}\}$. For an Open Curve, $\{\text{Triplets}\}$ are in the Interior only.

For a Closed Curve, $\{\text{Triplets}\}$ correspond to the entire Curve. (5.47)

6 Preliminary collection of data, by the Human.

Supposedly without an already given Parametrization in $\mathbb{R}^3$:

the Marks' Cartesian Coordinates are not an input, but a Curve's property that the Human's and Program's operations pull out. (6.1)

A preliminary work must be done: the Human must intrinsically define the Ensemble: Curve & Envelope, from the Trace. (6.2)

It is said "intrinsically". Intrinsic to a part of $\mathbb{R}^3$. At first, this part is the { Initial Reference Space }. This part grows up to the { Triangulation Nest }, that envelops the Ensemble at the end of operations. (6.3)

{ Initial Reference Space }. From the Marked Trace, a Curve ensues. For this Curve, an { Initial Reference Space } is conceived by the Human.

She chooses four, not all coplanar Marks on the Marked Trace. These Marks $\subset \{ \text{Marks} \}$ assume the surnames $\{ \text{Reference Marks} \} = \{ q0, q1, q2, q3 \}$. (6.4)

That is, since the four { Reference Marks } (say p_m, p_r, p_g, p_h) are important, each one of them has a surname: $p_m q0, p_r q1, p_g q2, p_h q3$. (6.5)

Not necessarily the { Reference Marks } are a continuous interval of the { Marks } sequence. (6.6)

In the { Initial Reference Space }, the { Reference Marks } are the specified, assumed end terminals of the Human's measurements in $\mathbb{R}^3$ of four straight Distances, for any Mark (p) of { Marks }. (6.7)

The { Reference Marks } are the vertexes of the, generally non-regular, **Reference Tetrahedron**. Obviously, not a degenerate Tetrahedron. (For a plane Trace, an out-of-plane Segment is adjoined to an end of the Trace, giving hospitality to one out-of-plane Reference Mark.) (6.8)

The Human has a metering stick; she intrinsically measures

{ Distances: {Marks}<->{Reference Marks} }:

Matrix $dv = \{ \forall p; \ ||p - q0||, ||p - q1||, ||p - q2||, ||p - q3|| \}$. (6.9)

Completing the Matrix $\{ \text{Indexed Marks; Marks' Distances} \} = dv$ the Human gets an Intrinsic Definition of the Array $\{ \text{Marks} \}$, relative to the Curve's $\{ \text{Initial Reference Space} \}$. (6.10)

For any specified Mark (p) of the Array $\{ \text{Marks} \}$, in general 4 non-null $\{ \text{Mark's Distances} \}_p$ correspond. 3 non-null and a null one if the examined Mark (p) coincides with one of $\{ q_0, q_1, q_2, q_3 \}$. (6.11)

Within the set $\{ \text{Distances} \}$, there are the six $\{ \text{Reference Distances among the Reference Marks} \} =$
 $\{ d_{01} = \| q_0 - q_1 \|, d_{12} = \| q_1 - q_2 \|, d_{20} = \| q_2 - q_0 \|, d_{03} = \| q_0 - q_3 \|, d_{13} = \| q_1 - q_3 \|, d_{23} = \| q_2 - q_3 \| \}$. (6.12)

Hints.

The subset $\{ \text{Reference Marks} \}$ is preferably chosen near the Curve's approximate center of gravity, aiming to the containment of the lengths of the set $= \{ \text{Marks' Distances} \} = \{ \text{Distances: } \{ \text{Marks} \} \leftrightarrow \{ \text{Reference Marks} \} \} \subset dv$. (Hint 1) (6.13)

It is advisable to choose far apart $\{ q_0, q_1, q_2, q_3 \}$: their spread increases precision of the measurement. (Hint 2) (6.14)

A substantial non-planarity of $\{ q_0, q_1, q_2, q_3 \}$ helps to get rid of the Marks' Mirror Images. (Hint 3) (6.15)

If the Curve's No. of Marks is N_M , then $Number_{Distances} \approx 4 N_M$. (6.16)

Each Mark's measurement is independent from the other Marks' measurements, not independent from the measurements of the considered Reference Mark, obviously. *There is not a succession of dependent measurements along the marked Trace.* (6.17)

The Human stores the Matrix $\{ \text{Indexed Marks; Marks' Distances} \} = dv$ in order to give it to CPlanes.py. (6.18)

Marked Trace's handedness: a left - handed Trace must be represented by a left - handed Curve, and a right - handed Trace by a right - handed Curve. (6.19)

Left and Right are ascertained on a chosen $\{ \text{Handedness Reference Interval} \}$

of the marked Trace, since, in general, the marked Trace may have left and right-handed parts in sequence, and so the ensuing Curve. (6.20)

The Human acknowledges the marked Trace's handedness and defines the corresponding parameter "Handedness". (6.21)

{ Reference Marks }, { Reference Distances } and "Handedness" are defining properties of the ensuing Ensemble. (6.22)

7 The Human and CPlanes. py

The Human communicates: {Reference Marks}, {Reference Distances}, {Indexed Marks ; Marks' Distances}, " Handedness " to the program CPlanes.py.

The 1st intervention of CPlanes. py. If the program CPlanes py is at the Human's disposal, it helps; CPlanes. py can determine:

The Cartesian axes: x, y, z across q0. (7.1)

The Reference Marks' coordinates:

q0(0, 0, 0); q1(d01, 0, 0); q2(x(q2), y(q2), 0); q3(x(q3), y(q3), z(q3)).

q1 lies on the positive x axis.

q2 lies in the x y plane with $y(q2) > 0$.

Handedness = "right" gives $z(q3) > 0$.

Handedness = "left" gives $z(q3) < 0$. (7.2)

By them, the Human gets a coordinated { Initial Reference Space } $\subset \mathbb{R}^3$.

It is the { Inital Reference Space } because it shall be implemented again, getting the { Triangulation Nest Reference Space } that envelops the Ensemble. (7.3)

The 2nd intervention of CPlanes. py. CPlanes.py effects the Reconstruction of any Mark $p \in dv$ by 4 Trilaterations by the 4 Triples { {q0, q1, q2}, {q0, q1, q3}, {q0, q2, q3}, {q2, q1, q3} } of the Tetrahedron. (7.4)

For each p, each Trilateration - one Triple - { three Distances } - yields two

Solutions: one, and its Mirror Image. (7.5)

For each p , the 4 Trilaterations yield 8 *Solutions*. (7.6)

For each p , the array $\{ \text{Candidates} \}_p$ stores the *Solutions*. (7.7)

By the array $\{ \text{accepted} \}$:

For each p , CPlanes.py does the Validation, i.e. the *selection*:
accepted $\vee$ inconsistent $\vee$ ambiguous *Position* p . (7.8)

For each p , CPlanes.py chooses the best Candidate Position, naming it the "reconstructed p ", gives its *Coordinates*, and discards the Mirror Images. (7.9)

For each p , it is sufficient one false Distance_p of 4 to contaminate 3 Distances_p and render suspect all 4 Distances_p . Result: a Human's new measurement of 4 Distances_p is needed. (7.10)

The above, $\forall p \in dv$. (7.11)

Hence CPlanes.py gives the Matrix $\{ \text{Coordinates} \}$ for dv . (7.12)

The above is distance-geometry with redundancy. It is over-determined by purpose. (7.13)

Notice. Said interventions of CPlanes.py have an argument: the set $\{ \text{Indexed Marks} \}$ of the Curve, not all $\{ \text{Points} \}$ of the Curve. The Curve's Arcs, and their - Arc-defined - $\{ \text{Extra Points} \}$ are not yet specified, hence not yet considered at this stage. By the above data it is not yet possible to draw sufficiently satisfying plots of the Curve. (7.14)

The 3rd intervention of CPlanes.py.

$\{ \text{Arcs} \}$, $\{ \text{Extra Points} \}$ are calculated. (7.15)

The 4th intervention of CPlanes.py: Plots.

CPlanes.py can produce for a Curve:

a] The Curve's Plot, with Indexed Marks, Curvature Centers, Boundary Vector Radii, Arc-based densification by Extra Points.

b] At any Mark, for the Outgoing Frenet Serret frame, with reference to the

Incoming Frenet Serret frame, 3 Plots:
T comparison, N comparison, B comparison. (7.16)

Planar Trace. A planar Trace is managed by adjoining one out-of-plane Segment to one of its ends.

Appendices

Appendix A The full Python source code for CPlanes class is here:

<https://github.com/giuse/trace/CPlanes.py>

Appendix B The Plot of bare Curve 8 and its 3 Plots: T comparison, N comparison, B comparison.

The Plots of: Curve 4, Curve 11 are available.

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